

MENSTRUAL CYCLE PREDICTION

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Abstract

- The Menstrual Cycle Prediction System is designed to predict the next menstrual period date based on user input such as last menstrual date, average cycle length, lifestyle factors, and health indicators.
- The project focuses on providing a simple and user-friendly interface for women to track and understand their menstrual health.
- Factors like stress level, sleep duration, and physical activity are considered to estimate possible cycle delays.
- The system also predicts ovulation and fertile windows to assist in reproductive health awareness.
- Health suggestions and pain relief measures are included to support users during menstruation.
- The application aims to improve awareness, reduce uncertainty, and promote healthy lifestyle practices.
- The system is developed using HTML, CSS, and JavaScript with data visualization for better understanding

Introduction

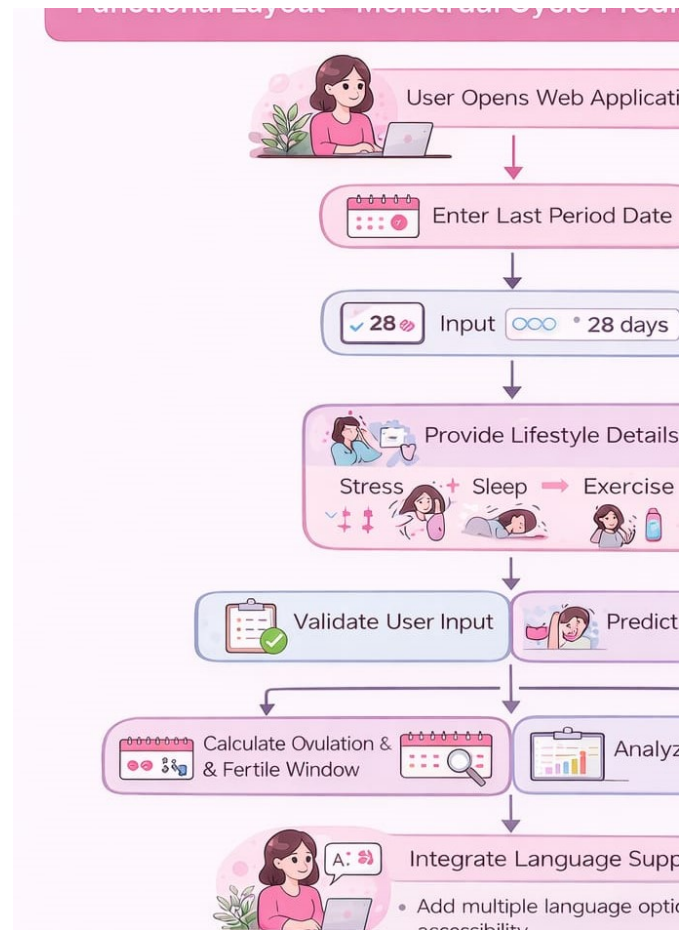
- Menstrual health plays a crucial role in the overall well-being of women.
- Irregular menstrual cycles may indicate stress, lifestyle imbalance, or underlying health conditions.
- Many women face difficulty in predicting their menstrual cycle accurately.
- This project introduces a Menstrual Cycle Prediction System to estimate upcoming cycle dates.
- The system uses user-provided inputs such as last period date and average cycle length.
- Lifestyle parameters like sleep, stress, and exercise are incorporated to improve prediction accuracy.
- The project provides ovulation and fertile window information for better reproductive planning.
- The system also offers basic health suggestions and pain relief tips during menstruation.
- This project is aimed at providing a digital solution for menstrual tracking and awareness.

Literature Survey

Ref. No	Title & Author	Technique Used	Pros	Cons
1	Menstrual Cycle Tracking Using Digital Health Tools – Smith et al.	Mobile-based tracking	Easy to use and accessible	Limited medical accuracy
2	Lifestyle Factors Affecting Menstrual Regularity – Johnson	Statistical analysis	Identifies key lifestyle effects	Requires large datasets
3	Ovulation Prediction Methods – Brown	Calendar-based method	Simple calculation	Less accurate for irregular cycles
4	Women Health Monitoring Systems – Lee et al	Rule-based systems	Easy implementation	Does not adapt automatically
5	Stress Impact on Menstrual Cycle – Kumar	Survey-based analysis	Highlights stress influence	Subjective data

Functional Layout

STRUAL CYCLE PREDICTION :



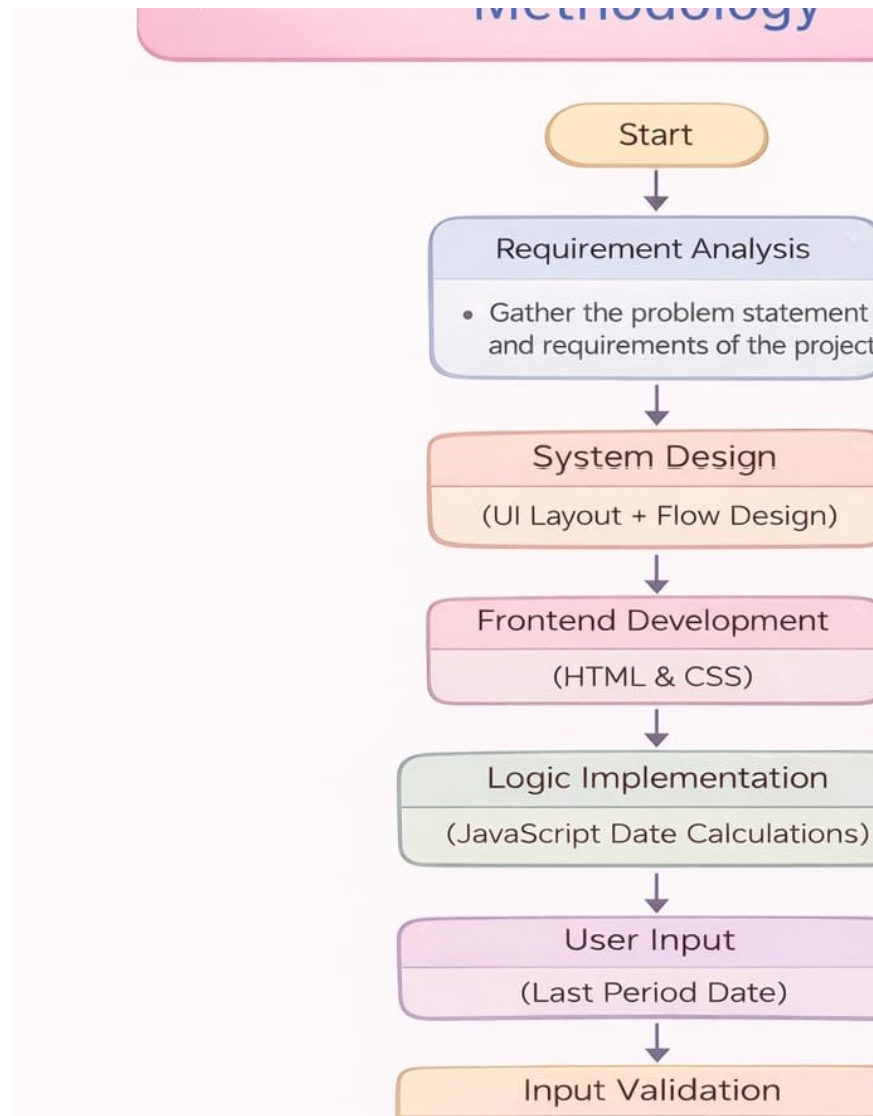
Objective

- To design a menstrual cycle prediction system that accurately estimates the next period date using user input.
- To provide ovulation, fertile window prediction, and health suggestions for improving menstrual awareness.

Outcomes

- The system successfully predicts the next menstrual cycle and fertile period using the user's last period date.
- A simple and user-friendly web interface allows easy data entry and clear result display.
- Automated cycle calculation reduces manual errors and improves accuracy.
- Input validation ensures reliable and correct prediction results.
- The modular design makes the system easy to maintain, enhance, and extend in the future.

Methodology



Implementation

Components

Frontend user interface using HTML and CSS
Logic processing using JavaScript
Chart module for data visualization

Specifications

User-friendly interface
Browser-based execution
Lightweight and responsive design

Coding

JavaScript functions for prediction calculations
Condition-based logic for delay estimation
Local storage for cycle history

➤ Application

- Menstrual cycle tracking
- Ovulation and fertile window prediction
- Period pain relief and health guidance

➤ Integration

- Integration of prediction logic with UI
- Integration of charts for data visualization
- Integration of stored data for cycle analysis
- The system integrates all modules to provide accurate predictions and health suggestions.
- The implementation ensures ease of use, reliability, and informative output for users.

Results/Working

The screenshot shows a web application interface for a 'Menstrual Cycle Prediction System'. The top navigation bar is dark with a hamburger menu icon, the text 'Menstrual Cycle ...', and a pencil icon. Below the navigation bar, there are two tabs: 'mcp.html' and 'Menstrual Cycl...'. The main content area has a light pink background with the title 'Menstrual Cycle Prediction System' in a bold, dark pink font. Below the title, there is a white form with the following fields: 'Average Cycle Length (days)' with a placeholder 'Eg: 28', 'Stress Level (1-5)' with a placeholder '1 = Low, 5 = High', 'Sleep Hours' with a placeholder 'Eg: 7', and 'Exercise Level (1-5)' with a placeholder '1 = Low, 5 = High'. At the bottom of the form is a large pink button labeled 'Predict'.

Menstrual Cycle ...

mcp.html Menstrual Cycl...

Menstrual Cycle Prediction System

Average Cycle Length (days)

Eg: 28

Stress Level (1-5)

1 = Low, 5 = High

Sleep Hours

Eg: 7

Exercise Level (1-5)

1 = Low, 5 = High

Predict

Results/Working

☰ Menstrual Cycle ... ✎

mcp.html

Menstrual Cycl...

30

Stress Level (1-5)

3

Sleep Hours

6

Exercise Level (1-5)

3

Predict

Prediction Result

Predicted Delay Days: 0

Expected Cycle Length: 30 days

Health Suggestions

Conclusion & Future Scope

➤ Conclusion

- The proposed system provides a simple and effective solution for menstrual cycle prediction.
- It helps users track periods, reduce anxiety, and prepare in advance.
- Pain-relief suggestions make the application health supportive, not just predictive.
- The application is lightweight, easy to use, and suitable for students and beginners.
- This project demonstrates practical usage of HTML, CSS, and JavaScript.

➤ Future Scope

- Allow users to customize cycle length instead of fixed 28 days.
- Add period history tracking with calendar view.
- Include ovulation and fertile window prediction.
- Integrate symptom analytics using AI/ML.
- Push notifications/reminders before the next period.
- Convert to a mobile app (Android/iOS).

References (APA Style)

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THANK YOU